Today

- Commutation relations
- Time-independent Schrödinger equation
- Begin topic 2
  - The infinite square well

1.13 Commutation relations between position and momentum

We have already seen that the position and the momentum operators are given by

$$\hat{\vec{r}} = \vec{r}, \qquad \hat{\vec{p}} = -i\hbar\vec{\nabla}. \tag{1.13.1}$$

Now consider applying $\hat{x}$ first and $\hat{p}_x$ second on the wavefunction $\psi(x,t)$,

$$\hat{p}_x\hat{x}\psi(x,t) = \hat{p}_x\left(\hat{x}\psi(x,t)\right) = -i\hbar\frac{\partial}{\partial x}\left(x\psi(x,t)\right) = -i\hbar\psi(x,t) - i\hbar x\frac{\partial}{\partial x}\psi(x,t). \tag{1.13.2}$$

Next, suppose we reverse the order of operations by applying $\hat{p}_x$ first and $\hat{x}$ second,

$$\hat{x}\hat{p}_x\psi(x,t) = \hat{x}\left(\hat{p}_x\psi(x,t)\right) = x\left(-i\hbar\frac{\partial}{\partial x}\psi(x,t)\right) = -i\hbar x\frac{\partial}{\partial x}\psi(x,t). \tag{1.13.3}$$

Thus, we see that $\hat{p}_x\hat{x}\psi(x,t) \neq \hat{x}\hat{p}_x\psi(x,t)$, and

$$\left(\hat{x}\hat{p}_x - \hat{p}_x\hat{x}\right)\psi(x,t) = i\hbar\,\psi(x,t) \tag{1.13.4}$$

Denoting $\hat{x}\hat{p}_x - \hat{p}_x\hat{x}$ by $[\hat{x},\,\hat{p}_x]$, called the commutator between $\hat{x}$ and $\hat{p}_x$,

$$[\hat{x},\,\hat{p}_x] = i\hbar. \tag{1.13.5}$$

Because $[\hat{x},\,\hat{p}_x] \neq 0$, we say $\hat{x}$ and $\hat{p}_x$ do not commute.

On the other hand,

$$[\hat{x},\,\hat{p}_y]\psi(\vec{r},t) \quad = \quad x(-i\hbar\frac{\partial}{\partial y})\psi(\vec{r},t) - (-i\hbar\frac{\partial}{\partial y}(x\psi(\vec{r},t)))$$

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$$= \quad x(-i\hbar\frac{\partial}{\partial y})\psi(\vec{r},t) + x(i\hbar\frac{\partial}{\partial y}(\psi(\vec{r},t))) = 0, \tag{1.13.6}$$

and $\hat{x}$ and $\hat{p}_y$ commute.

Similarly and collecting,

$$[\hat{x},\,\hat{p}_x] = [\hat{y},\,\hat{p}_y] = [\hat{z},\,\hat{p}_z] = i\hbar \tag{1.13.7}$$

and the components of $\hat{\vec{r}}$ and $\hat{\vec{p}}$ commute for any other combination, such as

$$[\hat{x},\,\hat{p}_y] = [\hat{y},\,\hat{p}_x] = [\hat{x},\,\hat{y}] = [\hat{p}_x,\,\hat{p}_y] = 0. \tag{1.13.8}$$

We will investigate later that when two operators do not commute there is uncertainty in measuring them simultaneously.

Denoting the components of the vectors by numerical indices with $x = x_1$, $y = x_2$, $z = x_3$, $p_x = p_1$, $p_y = p_2$, and $p_z = p_3$, the commutation relations among the components of $\hat{\vec{r}}$ and $\hat{\vec{p}}$ can be summarized as

$$[\hat{x}_i,\,\hat{p}_j] = i\hbar\delta_{ij}, \quad [\hat{x}_i,\,\hat{x}_j] = [\hat{p}_i,\,\hat{p}_j] = 0, \tag{1.13.9}$$

where δ_{ij} is the Kronecker delta defined as $\delta_{ij} = 1$ for $i = j$ and $\delta_{ij} = 0$ for $i \neq j$.

Exercise 1.13.1: Determine $[\hat{x},\,\hat{p}_x^2]$.

Solution: $[\hat{x},\,\hat{p}_x^2] = \hat{x}\hat{p}_x^2 - \hat{p}_x^2\hat{x} = \hat{x}\hat{p}_x\hat{p}_x - \hat{p}_x\hat{p}_x\hat{x} = \hat{x}\hat{p}_x\hat{p}_x - \hat{p}_x\hat{x}\hat{p}_x + \hat{p}_x\hat{x}\hat{p}_x - \hat{p}_x\hat{p}_x\hat{x} = [\hat{x},\,\hat{p}_x]\hat{p}_x + \hat{p}_x[\hat{x},\,\hat{p}_x] = i\hbar\hat{p}_x + \hat{p}_x i\hbar = 2i\hbar\hat{p}_x$.

In general, given three operators $\hat{A}$, $\hat{B}$, and $\hat{C}$, we have $[\hat{A},\,\hat{B}\hat{C}] = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{C}\hat{A} = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{A}\hat{C} + \hat{B}\hat{A}\hat{C} - \hat{B}\hat{C}\hat{A}$, where we have subtracted and added $\hat{B}\hat{A}\hat{C}$. Therefore, $[\hat{A},\,\hat{B}\hat{C}] = [\hat{A},\,\hat{B}]\hat{C} + \hat{B}[\hat{A},\,\hat{C}]$. Thus,

$$[\hat{A},\,\hat{B}\hat{C}] = \hat{B}[\hat{A},\,\hat{C}] + [\hat{A},\,\hat{B}]\hat{C}. \tag{1.13.10}$$

1.14 Time-independent Schrödinger equation

Suppose the potential energy is independent of time; i.e., $V(\vec{r},t) = V(\vec{r})$. The Schrödinger equation then becomes

$$i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t) + V(\vec{r})\psi(\vec{r},t), \tag{1.14.1}$$

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Because the potential no longer mixes position and time, we can use method of separation of variables and start with the ansatz

$$\psi(\vec{r},t) = \psi(\vec{r})\phi(t). \tag{1.14.2}$$

With this ansatz, (1.14.1) becomes

$$i\hbar\psi(\vec{r})\frac{d\phi(t)}{dt} = -\phi(t)\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r}) + V(\vec{r})\phi(t)\psi(\vec{r}). \tag{1.14.3}$$

Dividing both sides by $\psi(\vec{r})\phi(t)$,

$$i\hbar\frac{1}{\phi(t)}\frac{d\phi(t)}{dt} = \frac{1}{\psi(\vec{r})}\left[-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r}) + V(\vec{r})\right]. \tag{1.14.4}$$

Because the left hand side is a function of only t and the right hand side is a function of only $\vec{r}$, each side must be a constant. Let the constant be E. The equation of motion for $\phi(t)$ is then

$$i\hbar\frac{d\phi(t)}{dt} = E\phi(t), \tag{1.14.5}$$

and the solution is

$$\phi(t) = e^{-\frac{i}{\hbar}Et}, \tag{1.14.6}$$

where we have simply set constant coefficient to one, since any constant can be absorbed in $\psi(\vec{r})$.

The equation of motion of $\psi(\vec{r})$ is

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r}) + V(\vec{r})\psi(\vec{r}) = E\psi(\vec{r}), \tag{1.14.7}$$

or

$$\hat{H}\psi(\vec{r}) = E\psi(\vec{r}), \tag{1.14.8}$$

where $\hat{H} = -\frac{\hbar^2}{2m}\vec{\nabla}^2 + V(\vec{r})$ is the Hamiltonian operator. This is called the time-independent Schrödinger equation.

Observe that when the potential is time-independent, the time dependence of the wavefunction that corresponds to the constant E is easily written once we have solved the time-independent Schrödinger equation,

$$\psi_E(\vec{r},t) = \psi_E(\vec{r})e^{-\frac{i}{\hbar}Et}, \tag{1.14.9}$$

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where we have added subscript E to identify $\psi_E(\vec{r},t)$ as a wavefunction that corresponds to constant E.

Exercise 1.14.1: The wavefunction of a particle of mass m in a one-dimensional space is given to be $\psi(x,t) = Ae^{-\frac{x^2}{4\sigma^2}-\frac{i}{\hbar}Et}$, where A, σ, and E are real constants.

1. Write down the corresponding normalized wavefunction.
2. Determine the probability for finding the particle between $x = 0$ and $x = \infty$. (5 points)
3. Determine $\langle\hat{x}\rangle$, $\langle\hat{x}^2\rangle$, and $\Delta x \equiv \sqrt{\langle\hat{x}^2\rangle - \langle\hat{x}\rangle^2}$.
4. Determine the potential energy of the particle in terms of given parameters. Show that you obtain the standard simple harmonic potential $V(x) = \frac{1}{2}m\omega^2x^2$ for $E = \frac{1}{2}\hbar\omega$ and $\sigma = \sqrt{\frac{\hbar}{2m\omega}}$.

1.15 Properties of solutions for time-independent potentials

Since we will be investigating only time-independent potentials in nearly all of this course, it is important to summarize some important properties of the solutions when the potential is time-independent.

1.15.1 The constant E must be real

One way that the reality of E follows is from Born's hypothesis on the normalizability of the wavefunction. In particular, considering $\psi_E(\vec{r},t) = \psi_E(\vec{r})\, e^{-\frac{i}{\hbar}Et}$ and writing $E = E_R + iE_I$, where E_R and E_I are possible real and imaginary parts of E, we have $|\psi_E(\vec{r},t)|^2 = |\psi_E(\vec{r})|^2 e^{\frac{i}{\hbar}(E_R - iE_I)t}$ $e^{-\frac{i}{\hbar}(E_R+iE_I)t} = |\psi_E(\vec{r})|^2 e^{\frac{2}{\hbar}E_I t}$, and

$$\int |\psi_E(\vec{r},t)|^2 d^3\vec{r} = \int |\psi_E(\vec{r})|^2 d^3\vec{r} e^{\frac{2}{\hbar}E_I t}, \tag{1.15.1}$$

which cannot be set equal to one for all t except when $E_I = 0$. Therefore, E must be real.

1.15.2 Stationary wavefunction

Given the wavefunction $\psi_E(\vec{r},t)$ that corresponds to E,

$$\mathcal{P}(\vec{r},t) = |\psi_E(\vec{r},t)|^2 = |\psi_E(\vec{r})|^2, \tag{1.15.2}$$

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and the probability density is independent of time. We say the wavefunction $\psi_E(\vec{r}, t)$ is stationary.

Question 1.15.1: The uncertainty of the energy ΔE in a stationary state of energy E is (A). $\frac{\hbar}{2}$, (B). $\hbar$, (C). 0, (D). E, (E). $2E$.

1.15.3 Stationary wavefunction is a state of definite energy

Using $\hat{H}\psi_E(\vec{r}, t) = E\psi_E(\vec{r}, t)$, the expectation value of the Hamiltonian operator in state $\psi_E(\vec{r})$ is

$$\langle \hat{H} \rangle = \int \psi_E(\vec{r}, t)^* \hat{H} \psi_E(\vec{r}, t) d^3\hat{r} = \int \psi_E(\vec{r}, t)^* E \psi_E(\vec{r}, t) d^3\hat{r} = E \int |\psi_E(\vec{r}, t)^2 d^3\hat{r} = E. \quad (1.15.3)$$

Therefore, the constant E is the expectation value of the Hamiltonian operator, and it is energy. Furthermore,

$$\langle \hat{H}^2 \rangle = \int \psi_E(\vec{r}, t)^* \hat{H}^2 \psi_E(\vec{r}, t) d^3\hat{r} = \int \psi_E(\vec{r}, t)^* EE \psi_E(\vec{r}, t) d^3\hat{r} = E^2 \int |\psi_E(\vec{r}, t)^2 d^3\hat{r} = E^2, \quad (1.15.4)$$

and the uncertainty or variance of the energy is

$$\Delta E = \sqrt{\langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2} = \sqrt{E^2 - E^2} = 0. \quad (1.15.5)$$

Therefore, the stationary state $\psi_E(\vec{r}, t)$ is a state of definite energy E.

1.15.4 Spatial part of stationary wavefunction can be chosen to be real

Given $\psi_E(\vec{r})$ is a solution to the time-independent Schrodinger equation with energy E

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2 \psi_E(\vec{r}) + V(\vec{r})\psi_E(\vec{r}) = E\psi_E(\vec{r}), \quad (1.15.6)$$

and complex conjugating it, we have

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2 \psi_E(\vec{r})^* + V(\vec{r})\psi_E(\vec{r})^* = E\psi_E(\vec{r})^*. \quad (1.15.7)$$

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In other words, $\psi_E(\vec{r})^*$ is also a solution with the same energy. Therefore, we can trade $\psi_E(\vec{r})$ and $\psi_E(\vec{r})^*$ for two other combinations,

$$\psi_E(\vec{r}) + \psi_E(\vec{r})^* \qquad \text{and} \qquad \frac{\psi_E(\vec{r}) - \psi_E(\vec{r})^*}{i}, \tag{1.15.8}$$

which are both real. Thus, the spatial part of a stationary wavefunction can be chosen to be real.

1.15.5 Parity symmetry: Even and odd wavefunctions

Suppose the potential is symmetric under reflection about the origin, $V(-\vec{r}) = V(\vec{r})$. In this case, given

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_E(\vec{r}) + V(\vec{r})\psi_E(\vec{r}) = E\psi_E(\vec{r}), \tag{1.15.9}$$

and replacing $\vec{r}$ by $-\vec{r}$ and setting $V(-\vec{r}) = V(\vec{r})$, we have a

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_E(-\vec{r}) + V(\vec{r})\psi_E(-\vec{r}) = E\psi_E(-\vec{r}). \tag{1.15.10}$$

Therefore, $\psi_E(-\vec{r})$ is a solution with the same energy. We can trade $\psi_E(\vec{r})$ and $\psi_E(-\vec{r})$ for two other combinations,

$$\psi_E(\vec{r}) + \psi_E(-\vec{r}) \qquad \text{and} \qquad \psi_E(\vec{r}) - \psi_E(-\vec{r}). \tag{1.15.11}$$

Now $\psi_E(\vec{r}) + \psi_E(-\vec{r})$ is even under reflection while $\psi_E(\vec{r}) - \psi_E(-\vec{r})$ is odd. Thus, when the potential is symmetric under reflection through the origin, we can choose the stationary states to be either even or odd under reflection.

1.15.6 Orthogonality/orthonormality

Suppose $\psi_n(\vec{r}, t)$ is a solution with energy E_n and $\psi_m(\vec{r}, t)$ is a solution with energy E_m; i.e.,

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_n(\vec{r}) + V(\vec{r})\psi_n(\vec{r}) = E_n\psi_n(\vec{r}) \quad \text{and} \quad -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi_m(\vec{r})^* + V(\vec{r})\psi_m(\vec{r})^* = E_m\psi_m(\vec{r})^*, \tag{1.15.12}$$

where we have taken the complex conjugate for the second. Multiplying both sides of the first by $\psi_m(\vec{r})^*$ and the second by $\psi_n(\vec{r})$,

$$-\frac{\hbar^2}{2m}\psi_m(\vec{r})^*\vec{\nabla}^2\psi_n(\vec{r}) + V(\vec{r})\psi_m(\vec{r})^*\psi_n(\vec{r}) = E_n\psi_m(\vec{r})^*\psi_n(\vec{r})$$

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$$-\frac{\hbar^2}{2m}\psi_n(\vec{r})\vec{\nabla}^2\psi_m(\vec{r})^* + V(\vec{r})\psi_n(\vec{r})\psi_m(\vec{r})^* = E_m\psi_n(\vec{r})\psi_m(\vec{r})^* \tag{1.15.13}$$

Subtracting the two,

$$-\frac{\hbar^2}{2m}\left(\psi_m(\vec{r})^*\vec{\nabla}^2\psi_n(\vec{r}) - \psi_n(\vec{r})\vec{\nabla}^2\psi_m(\vec{r})^*\right) = (E_n - E_m)\psi_m(\vec{r})^*\psi_n(\vec{r}) \tag{1.15.14}$$

Recalling our computation and definition of the probability current, the left hand side equals $-i\hbar\vec{\nabla}\cdot\vec{J}$. Therefore,

$$-i\hbar\vec{\nabla}\cdot\vec{J} = (E_n - E_m)\psi_m(\vec{r})^*\psi_n(\vec{r}). \tag{1.15.15}$$

Integrating both sides, and using the divergence theorem, $\int\vec{\nabla}\cdot\vec{J}d^3\vec{r} = \oint\vec{J}\cdot d\vec{a}$,

$$-i\hbar\oint\vec{J}\cdot d\vec{a} = (E_n - E_m)\int\psi_m(\vec{r})^*\psi_n(\vec{r})d^3\vec{r}. \tag{1.15.16}$$

Now, taking the boundary of the area integral to infinity, the current vanishes, since a normalizable wavefunction and the corresponding current must vanish at infinity, and the left hand side vanishes. We then have

$$(E_n - E_m)\int\psi_m(\vec{r})^*\psi_n(\vec{r})d^3\vec{r} = 0. \tag{1.15.17}$$

We have two cases.

1. Case 1: $n = m$ and $E_n = E_m$. In this case, $E_n - E_m = 0$ and for a normalized wavefunction $\psi_n(\vec{r})$, $\int\psi_n(\vec{r})^*\psi_n(\vec{r})d^3\vec{r} = 1$.
2. Case 2: $n \neq m$ and $E_n \neq E_m$. In this case, $E_n - E_m \neq 0$ and $\int\psi_m(\vec{r})^*\psi_n(\vec{r})d^3\vec{r} = 0$. We say $\psi_n(\vec{r})$ and $\psi_m(\vec{r})$ are orthogonal.

Combining the two cases, we have the orthonormality relation

$$\int\psi_m(\vec{r})^*\psi_n(\vec{r})d^3\vec{r} = \delta_{mn}, \tag{1.15.18}$$

where

$$\delta_{mn} = \begin{cases} 1, & m = n \\ 0, & m \neq n \end{cases}$$

is called the Kronecker delta.

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1.15.7 Linearity

The Schrödinger equation is linear. Given stationary states $\psi_n(\vec{r},t) = \psi_n(\vec{r})e^{-\frac{i}{\hbar}E_n t}$, $n = 1, 2, ...$, a linear combination of these

$$\psi(\vec{r},t) = \sum_n c_n\psi_n(\vec{r},t) = \sum_n c_n\psi_n(\vec{r})e^{-\frac{i}{\hbar}E_n t}, \tag{1.15.19}$$

where c_n are complex constants, is a solution to the time-dependent Schrödinger equation $i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = \hat{H}\psi(\vec{r},t)$. To check that, the left hand side is $i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = \sum_n c_n\psi_n(\vec{r})i\hbar\frac{\partial}{\partial t}e^{-\frac{i}{\hbar}E_n t} = \sum_n c_n E_n\psi_n(\vec{r})e^{-\frac{i}{\hbar}E_n t}$, and the right hand side is $\hat{H}\psi(\vec{r},t) = \sum_n c_n\hat{H}\psi_n(\vec{r})e^{-\frac{i}{\hbar}E_n t} = \sum_n c_n E_n\psi_n(\vec{r})e^{-\frac{i}{\hbar}E_n t}$, and they are equal.

1.15.8 Time evolution

> **Question 1.15.2:** Given stationary states ψ_1 and ψ_2 with different energies, is $c_1\psi_1 + c\psi_2$ stationary? (A). Yes, (B). No.

Because $\psi(\vec{r},t) = \sum_n c_n\psi_n(\vec{r},t) = \sum_n c_n\psi_n(\vec{r})e^{-\frac{i}{\hbar}E_n t}$ with constant coefficients c_n is a solution to the time-dependent Schrödinger equation, and $\psi(\vec{r},0) = \sum_n c_n\psi_n(\vec{r},0) = \sum_n c_n\psi_n(\vec{r})$ at time $t = 0$, we note that given

$$\psi(\vec{r},0) = \sum_n c_n\psi_n(\vec{r},0) = \sum_n c_n\psi_n(\vec{r}) \tag{1.15.20}$$

at time $t = 0$, the time evolution, or the wavefunction at time t, is simply

$$\psi(\vec{r},t) = \sum_n c_n\psi_n(\vec{r},t) = \sum_n c_n\psi_n(\vec{r})e^{-\frac{i}{\hbar}E_n t}. \tag{1.15.21}$$

In other words, once we have expanded a given initial wavefunction in terms of a linear combination of the stationary wavefunctions $\psi_n(\vec{r})$, the time evolution follows simply by putting a factor of $e^{-\frac{i}{\hbar}E_n t}$ to each term containing $\psi_n(\vec{r})$.

1.15.9 Normalization

Given $\psi(\vec{r},t) = \sum_n c_n\psi_n(\vec{r},t) = \sum_n c_n\psi_n(\vec{r})e^{-\frac{i}{\hbar}E_n t}$ with complex coefficients c_n, normalizing it leads to

$$1 = \int |\psi(\vec{r},t)|^2 d^3\vec{r} = \int \left(\sum_m c_m^*\psi_m^*(\vec{r})e^{\frac{i}{\hbar}E_m t}\right)\left(\sum_n \psi_n(\vec{r})e^{-\frac{i}{\hbar}E_n t}\right) d^3\vec{r}$$

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$$= \sum_n \sum_m c_m^* c_n e^{\frac{i}{\hbar}(E_m - E_n)t} \int \psi_m^*(\vec{r})\psi_n(\vec{r}) d^3\vec{r} = \sum_n \sum_m c_m^* c_n e^{\frac{i}{\hbar}(E_m - E_n)t} \delta_{mn} = \sum_n |c_n|^2. \quad (1.15.22)$$

Therefore, the constant coefficients are constrained by

$$\sum_n |c_n|^2 = 1. \quad (1.15.23)$$

1.15.10 Completeness

Given an arbitrary $\psi(\vec{r}, t)$ that is a solution to the Schrödinger equation with certain boundary conditions, we can express it as a linear superposition of the stationary solutions $\psi_n(\vec{r}, t) = \psi_n(\vec{r}) e^{-\frac{i}{\hbar}E_n t}$ as

$$\psi(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}, t) = \sum_n c_n \psi_n(\vec{r}) e^{-\frac{i}{\hbar}E_n t}. \quad (1.15.24)$$

with complex coefficients c_n. In other words, the stationary orthonormal solutions form a complete set.

To find the condition for the above statement to hold, first let us find the coefficients c_n. Multiplying both sides by $\psi_m(\vec{r}, t)^*$ and integrating over $d^3\vec{r}$,

$$\int \psi_m(\vec{r}, t)^* \psi(\vec{r}, t) d^3\vec{r} = \sum_n c_n \int \psi_m(\vec{r}, t)^* \psi_n(\vec{r}, t) d^3\vec{r} = \sum_n c_n \delta_{mn} = c_m. \quad (1.15.25)$$

Therefore,

$$c_n = \int \psi_n(\vec{r}, t)^* \psi(\vec{r}, t) d^3\vec{r}. \quad (1.15.26)$$

Inserting these coefficients back into the expansion,

$$\begin{aligned} \psi(\vec{r}, t) &= \sum_n \int \psi_n(\vec{r}', t)^* \psi(\vec{r}', t) d^3\vec{r}' \psi_n(\vec{r}, t) \\ &= \int d^3\vec{r}' \psi(\vec{r}', t) \sum_n \psi_n(\vec{r}', t)^* \psi_n(\vec{r}, t). \end{aligned} \quad (1.15.27)$$

The stationary states form a complete set if

$$\sum_n \psi_n(\vec{r}', t)^* \psi_n(\vec{r}, t) = \delta^3(\vec{r}' - \vec{r}), \quad (1.15.28)$$

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where $\delta^3(\vec{r}' - \vec{r})$ is the Dirac-delta function in three dimensions defined by

$$\int d^3\vec{r}'\psi(\vec{r}', t)\delta^3(\vec{r}' - \vec{r}) = \psi(\vec{r}, t). \tag{1.15.29}$$

In other words, if we multiply and sum the stationary states as $\sum_n \psi_n(\vec{r}', t)^*\psi_n(\vec{r}, t)$ and we get $\delta^3(\vec{r}' - \vec{r})$, we know we have a complete set. Conversely, if we miss a state in the sum, we do not get $\delta^3(\vec{r}' - \vec{r})$. In cartesian coordinates, $\delta^3(\vec{r}' - \vec{r}) = \delta(x - x')\delta(y - y')\delta(z - z')$.

1.16 Delta function

Below, we summarize some of the properties of the delta function, focusing on one dimension. Given a function $f(x)$, the delta function $\delta(x - x')$ is defined such that

$$f(x') = \int f(x)\delta(x - x')dx \tag{1.16.1}$$

In other words, $\delta(x - x')$ in the integral projects out the value of $f(x)$ at the zero of the argument $x - x'$ of $\delta(x - x')$. Therefore, we can view $\delta(x - x')$ as defined by

$$\delta(x - x') = 0 \text{ for } x \neq x', \qquad f(x') = \int_a^b f(x)\delta(x - x')dx \text{ for } a < x' \text{ and } b > x'. \tag{1.16.2}$$

Because $\delta(x - x') = 0$ for $x \neq x'$, we can rewrite $f(x') = \int f(x)\delta(x - x')dx = f(x')\int \delta(x - x')dx$. Therefore,

$$\int \delta(x - x')dx = 1. \tag{1.16.3}$$

However, the delta function is not actually a function but a distribution. For instance, we can not interpret $\int \delta(x - x')dx = 1$ as the area under $\delta(x - x')$ being one.

Here we list some representations of the delta function,

$$\delta_\alpha(x) = \frac{1}{\alpha\sqrt{\pi}}e^{-\frac{x^2}{\alpha^2}} \tag{1.16.4}$$

$$\delta_\alpha(x) = \frac{1}{\pi x}\sin\left(\frac{x}{\alpha}\right) \tag{1.16.5}$$

$$\delta_\alpha(x) = \frac{\alpha}{\pi x^2}\sin^2\left(\frac{x}{\alpha}\right) \tag{1.16.6}$$

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With any of these representations, the delta function $\delta_\alpha(x-a)$ is defined through

$$f(x') = \lim_{\alpha\to 0}\int f(x)\delta_\alpha(x-x')dx. \tag{1.16.7}$$

However, $\lim_{\alpha\to 0}\int f(x)\delta_\alpha(x-x')dx \neq \int \lim_{\alpha\to 0} f(x)\delta_\alpha(x-x')dx$. Indeed the integral vanishes if the limit is taken within the integral.

Exercise 1.16.1: Show the following.

1. $\delta(ax) = \frac{1}{|a|}\delta(x)$
2. $\delta(x^2-a^2) = \frac{1}{2|a|}(\delta(x+a)+\delta(x-a))$
3. $\delta(f(x)) = \sum_i \frac{\delta(x-x_i)}{|f'(x)|}$, where x_i are the roots of $f(x)=0$.
4. Write $\delta^3(\vec{r}-\vec{r}')$ in spherical coordinates.
5. Write $\delta^3(\vec{r}-\vec{r}')$ in cylindrical coordinates.

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2 One-dimensional systems

Now, we begin constructing the wavefunctons for a system containing a single particle of mass m in one dimension. Different systems correspond to different potentials. We will be considering only potentials that are time-independent. The wavefunctions and the corresponding energies are then obtained by solving the time-independent Schrödinger equation in one dimension,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x). \tag{2.0.1}$$

Once we find the wavefunction $\psi_E(x)$ of energy E, its time dependence follows easily,

$$\psi_E(x,t) = \psi_E(x)e^{-\frac{i}{\hbar}Et}. \tag{2.0.2}$$

2.1 Boundary conditions

The boundary conditions we impose depend on the potential. Suppose we are given a generic potential $V(x)$. Consider a point $x = a$ on the $x-$axis. We want to investigate whether the wavefunction $\psi(x)$ and its first derivative $\psi'(x)$ are continuous or not across $x = a$. First, rewriting the Schrödnger equation as

$$\psi''(x) = \frac{2m}{\hbar^2}(V(x) - E)\psi(x), \tag{2.1.1}$$

we note that the continuity or discontinuity of $\psi''(x)$ follows the continuity or discontinuity of the potential $V(x)$.

For the first derivative, and considering a particle of finite energy E,

$$\begin{aligned}\lim_{\epsilon\to 0}(\psi'(a+\epsilon) - \psi'(a-\epsilon)) &= \lim_{\epsilon\to 0}\int_{a-\epsilon}^{a+\epsilon}\psi''(x)dx = \lim_{\epsilon\to 0}\int_{a-\epsilon}^{a+\epsilon}\frac{2m}{\hbar^2}(V(x)-E)\psi(x)dx \\ &= \frac{2m}{\hbar^2}\int_{a-\epsilon}^{a+\epsilon}V(x)\psi(x)dx,\end{aligned} \tag{2.1.2}$$

Now if $V(x)$ has a finite jump at $x = a$, this integral vanishes and $\psi'(x)$ is continuous across $x = a$. On the other hand, if $V(x)$ has an infinite jump, similar to the delta function, the integral will be nonzero but finite, and $\psi'(x)$ is discontinuous across $x = a$.

For the wavefunction itself,

$$\lim_{\epsilon\to 0}(\psi(a+\epsilon) - \psi(a-\epsilon)) = \lim_{\epsilon\to 0}\int_{a-\epsilon}^{a+\epsilon}\psi'(x)dx. \tag{2.1.3}$$

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Now whether $V(x)$ has a finite or infinite jump at $x = a$, this integral vanishes, since $\psi'(x)$ could have only a finite jump across $x = a$. Therefore, $\psi(x)$ is always continuous across $x = a$.

We summarize the boundary conditions as follows

1. When the potential $V(x)$ has a finite jump at $x = a$, both $\psi(x)$ and $\psi'(x)$ are continuous across $x = a$.
2. When the potential $V(x)$ has an infinite jump at $x = a$, $\psi(x)$ is continuous but $\psi'(x)$ is discontinuous across $x = a$.

Furthermore, the wavefunction must vanish at $x = \pm\infty$ for a normalizable wavefunction.

2.2 Infinite square well

Consider a particle of mass m in the potential

$$V(x) = \begin{cases} 0, & 0 \le x \le a \\ \infty, & x < 0 \text{ and } x > a. \end{cases}$$

The potential is shown in the figure below.

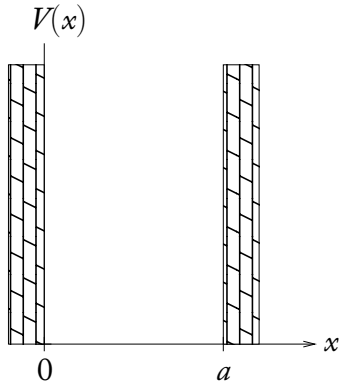


Figure 2.2.1: Infinite square well potential.

Because, the particle cannot have infinite energy, it cannot be found in the regions $x < 0$ and $x > a$. Therefore,

$$\psi(x) = 0, \quad \text{for } x < 0 \text{and } x > a. \tag{2.2.1}$$

For the region in the well, $0 \le x \le a$, we have $V(x) = 0$ and the Schrödinger equation reduces

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to

$$\psi''(x) + k^2\psi(x) = 0, \tag{2.2.2}$$

where

$$k = \sqrt{\frac{2mE}{\hbar^2}} \tag{2.2.3}$$

and $E > 0$. We can write the general solution in terms of linear combinations of real solutions $\sin kx$ and $\cos kx$ as

$$\psi(x) = A\sin kx + B\cos kx. \tag{2.2.4}$$

Applying the boundary condition on the continuity of $\psi(x)$ at $x = 0$ sets $\psi(0) = B = 0$. We then have $\psi(x) = A\sin kx$, and applying the boundary condition at $x = a$ sets $\psi(a) = A\sin ka = 0$. Therefore, either $A = 0$ or $ka = n\pi$, $n = 1, 2, ...$ But $A = 0$ sets $\psi(x) = 0$ everywhere, which corresponds to no particle anywhere. Therefore,

$$k_n = \frac{n\pi}{a}, \quad n = 1, 2, ... \tag{2.2.5}$$

The wave vector is discrete, which means that both the momentum and the energy of the particle are discrete too, since $p = \hbar k$ and $E = \sqrt{\frac{2mE}{\hbar^2}}$. Therefore, $\psi_n(x) = A_n \sin k_n x = A_n \sin(\frac{n\pi}{a}x)$. Normalizing and choosing the wavefunctions to be real, $1 = \int_0^a |\psi_n(x)|^2 dx = \int_0^a A_n^2 \sin^2(\frac{n\pi}{a}x)dx = A_n^2\frac{a}{2}$, or $A_n = \sqrt{\frac{2}{a}}$. The corresponding energies are $E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2\pi^2 n^2}{2ma^2}$. Summarizing, the normalized wavefunctions and the corresponding energies for the particle of mass m in the given infinite square well are

$$\psi_n(x) = \sqrt{\frac{2}{a}}\sin(\frac{n\pi}{a}x), \quad E_n = \frac{\hbar^2\pi^2 n^2}{2ma^2}, \qquad n = 1, 2, 3, ... \tag{2.2.6}$$

The wavefunctions and the corresponding energies are shown in the schematic figure below.

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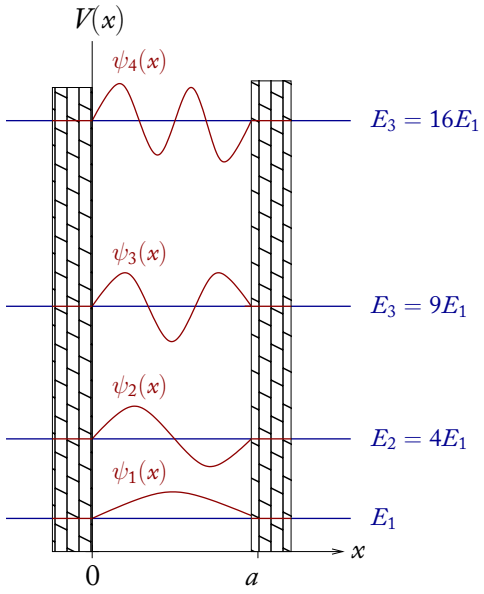


Figure 2.2.2: Wavefunctions and energies for a particle in the infinite square well potential.

Question 2.2.1: For the wavefunctions of the infinite square well, (A). $\psi(x)$ is continuous, (D). $\psi'(x)$ is continuous, (C). $\psi''(x)$ is continuous.

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